

is measured by these four things in order named: perseverance, careful methods of analysis, intuition, [and] luck.”

Common Cryptanalysis techniques

Coding “secret” information so that it is accessible only to the people you want it visible to is a practice that dates back centuries. And as long as there have been codes, there have been people looking to break them, for reasons ranging from mere academic interest to stopping potential global Armageddon (Enigma and World War II ring any bells?). As is a natural consequence, over the years many different and varied methods of cryptanalysis have been invented and discovered. Most of the current in-vogue methods for breaking modern cryptosystems rely heavily on the use of pure mathematics to solve problems, making them quite complex, and hence beyond the purview of this brief guide.

Instead, we will look at some of the more common cryptanalysis methods that can be used to crack at least the most basic codes. NOTE: These decryption methods are very elementary, and unlikely to be able to crack any of the codes commonly used today. So, restrict using these methods to breaking the code used by your younger brother/sister and watching their astonished expressions as you reveal the information they thought safe from your prying eyes. DISCLAIMER: This writer does not take any responsibility for sibling rivalries/angst created from the abovementioned incident. Use these methods at your own risk!

Frequency distributions

One of the most elementary forms of cryptanalysis uses something called “frequency distribution”. Essentially, this is a tendency of language where certain characteristics of a language stand out noticeably, in ciphertext as well as plaintext. Someone who has knowledge of these characteristics can use them to break a code in that language quite easily.

For example, in the English language, the letter ‘E’ is by far the most used letter of the alphabet. This means that its occurrence in any message text is mostly the highest. Therefore, when we see a letter being repeated very often, it is safe to assume that this is a replacement of the letter ‘E’. Using this as the starting point, we can discover other letter substitutions and accordingly “crack the code”. Similarly, we can use the common occurrence of the digraphic “th” as a starting point for solving most common codes.